3.5 π^e or e^π?

WORKSHEET

The question

STEP II 2013, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q1

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Taking logarithms, $a^b = b^a$ is the same as $b \ln a = a \ln b$. Rearrange this so that a appears only on one side and b only on the other.

STAGE 1 The tangent through the origin

Covers: Part (i)

Ask yourself If $y = mx$ touches $y = \ln x$ at $x = x_0$, what two equations must hold?

Pause and try If $y = mx$ cuts the curve at $x = a$ and $x = b$, write down $\ln a$ and $\ln b$ in terms of m. Eliminate m, then exponentiate.

STAGE 2 Lines that cut the y-axis above the origin

Covers: Part (ii)

Pause and try Now $\ln p = mp + c$ and $\ln q = mq + c$ with $c > 0$. Write $\frac{\ln p}{p}$ and $\frac{\ln q}{q}$ in terms of m and c, and compare them.

STAGE 3 Chords to the right of e

Covers: Part (iii)

Ask yourself The chord from $(p, \ln p)$ to $(q, \ln q)$, with $e \leq p < q$, is less steep than the curve at p. Why? What does that tell you about the chord's y-intercept?

Pause and try Use this and part (ii) with $p = e$ and $q = \pi$.

STAGE 4 Chords parallel to the tangent

Covers: Part (iv)

Pause and try The chord is parallel to the tangent $y = \frac{x}{e}$, and the curve lies below that tangent. Does the chord cut the y-axis above or below the origin? Then repeat part (ii) with the sign of c reversed.

What's really going on

The graph of $\frac{\ln x}{x}$

Everything in this question is visible in the graph of $h(x) = \frac{\ln x}{x}$ (Figure 3.5.1). Its derivative is $h'(x) = \frac{1-\ln x}{x^2}$, so h increases on $(0, e)$, reaches its maximum $\frac{1}{e}$ at $x = e$, and then decreases towards 0. A horizontal line at height m between 0 and $\frac{1}{e}$ cuts the graph twice, at points $a < e < b$ with $h(a) = h(b)$, which is exactly $a^b = b^a$. To the right of e, h is decreasing, so for $e \leq p < q$ we have $h(p) > h(q)$, that is, $p^q > q^p$. In particular $e^\pi > \pi^e$, and more generally $e^x > x^e$ for every $x \neq e$: no other number beats e at this game.

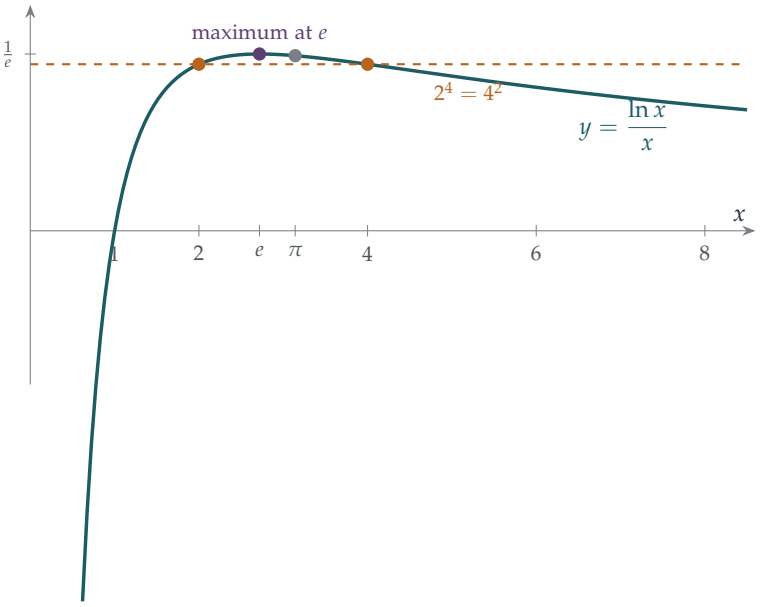


Figure 3.5.1 The graph of $h(x) = \frac{\ln x}{x}$. It rises to a maximum of $\frac{1}{e}$ at $x = e$ and then falls. Because $h(e) > h(\pi)$, we get $e^\pi > \pi^e$. The horizontal line at height $\frac{\ln 2}{2}$ (amber) meets the graph at $x = 2$ and $x = 4$: the only pair of distinct whole numbers with $n^m = m^n$.

All solutions of $x^y = y^x$

Which pairs of distinct positive reals satisfy $x^y = y^x$? A neat trick finds them all. Put $y = tx$ with $t > 1$; then $x^{tx} = (tx)^x$ gives $x^t = tx$, so $x^{t-1} = t$ and

$$x = t^{1/(t-1)}, \qquad y = t^{t/(t-1)}.$$

Writing $t = 1 + \frac{1}{n}$ gives $x = \left(1 + \frac{1}{n}\right)^n$ and $y = \left(1 + \frac{1}{n}\right)^{n+1}$. With $n = 1$ this is $(2, 4)$; with $n = 2$ it is $\left(\frac{9}{4}, \frac{27}{8}\right)$, and indeed $\left(\frac{9}{4}\right)^{27/8} = \left(\frac{27}{8}\right)^{9/4}$. As $n \to \infty$ both tend to e, the point where the two branches meet. The problem was discussed in letters between Daniel Bernoulli and Christian Goldbach in 1728, and later by Euler.

The most economical base

The function $x^{1/x} = e^{h(x)}$ is also maximised at $x = e$. This answers a curious question: if the "cost" of a number system is the base times the number of digits needed, the most economical base is e, and among whole numbers, 3 (narrowly beating 2). A few experimental computers, such as the Soviet Setun of 1958, have used base 3.

Try this next

1 **Whole numbers.** Try STEP I 1988, Question 1 (step.maths.org/questions/88-s1-q1).

2 **More comparisons.** Try STEP II 2025, Question 3 (in OCR's download of the 2025 papers, www.ocr.org.uk/Images/750486-2025-past-papers.zip). It uses the graph from this section to settle more comparisons between a^b and b^a, including one where a numerical bound on e does the work.

3 **A one-line proof.** Use $e^x > 1 + x$ for $x \neq 0$, with $x = \frac{\pi}{e} - 1$, to prove $e^\pi > \pi^e$ directly.

Further reading

- **Eli Maor, *e: The Story of a Number*** (Princeton University Press). The history of e, including the equation $x^y = y^x$.
- **Paul Nahin, *An Imaginary Tale*** (Princeton University Press). Not only about $\sqrt{-1}$: full of curious calculations of exactly this kind.
- **Michael Spivak, *Calculus*** (Cambridge University Press). Chapter 11 covers the use of derivatives to understand the shape of graphs with complete rigour.

The full guided solution is in Section 3.5:
https://beyondstep.keithmcnulty.org/ch03/pi-e-or-e-pi.html